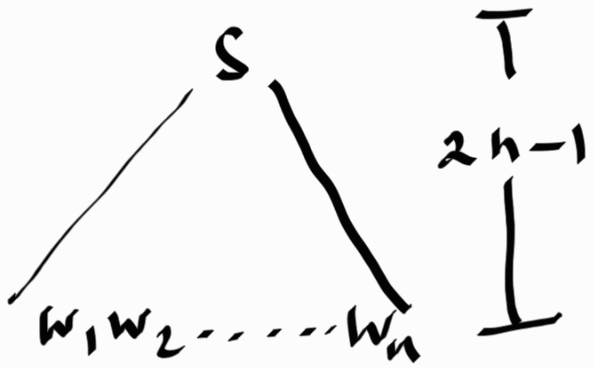
$A_{CFG} = \{ \langle G, w \rangle \mid G \text{ is a CFG and } G \text{ generates } w \}$

- Solution that we used:

$G \longrightarrow$ Convert G to CNF

Theorem 4.7



$s = w_1 w_2 \ldots\ldots w_n$

$O(|G|^{2n-1})$

G is a CFG in CNF

This is too costly.

- Solution 02
  - Dynamic programming Solution

If $A \Rightarrow BC$ is a rule and $B \overset{*}{\Rightarrow} X$ and $C \overset{*}{\Rightarrow} Y$ then $A \overset{*}{\Rightarrow} XY$

DP solution is based on this observation.

- Define a table[i, j] — which stores the collection of variables that generates the substring $w_i w_{i+1} \cdots w_j$

- Idea is simillar to matrix multiplication.

$$\underbrace{w_i w_{i+1} \cdots w_k}_{\text{in the table}[i,k] \text{ you have the variable } B} \Big| \underbrace{w_{k+1} \cdots w_j}_{\text{in the table you have variable } C}$$

Then If I have a rule $A \to BC$

then I can add A to table[i, j]

Refer the handout for the algorithm

Example

$$\begin{array}{l} \quad\ 1\ 2\ 3\ 4\ 5 \\ w = a\ a\ b\ c\ c \end{array}$$

$G:$
$$S \to AB$$
$$A \to AB \mid AA \mid a$$
$$B \to BC \mid b$$
$$C \to c$$

$m[1,2]$

$\overset{1}{a} \mid \overset{2}{a}$

$A \quad A$

$* \to AA$

$A \to AA$

i \ j	1	2	3	4	5
1	A	A			
2	–	A	S, A		
3	–	–	B	B	
4	–	–	–	C	$\emptyset$
5	–	–	–	–	C

$m[2,3]$

$\overset{2}{a} \mid \overset{3}{b}$

$A \quad B$

$* \to AB$

$m[3,4]$

$\overset{3}{b} \mid \overset{4}{c}$

$B \quad C$

$* \to BC$

$B \to BC$

$m[4,5]$

$\overset{4}{c} \mid \overset{4}{c}$

$C \quad C$

$* \to CC$

G : $S \to AB$

$A \to AB \mid AA \mid a$

$B \to BC \mid b$

$C \to c$

$w = aabcc$ (positions 1 2 3 4 5)

i \ j	1	2	3	4	5
1	A	A	A,S		
2	—	A	A,S	S,A	
3	—	—	B	B	B
4	—	—	—	C	ϕ
5	—	—	—	—	C

m[1,3]

a a b (1 2 3)

a | ab → A A,S

aa | b → A B

$* \to AA$

$\to AS$

$* \to AB$

m[2,4]

a b c

a | bc (2 | 3 4) → A B

ab | c (2 3 | 4) → A,S C

$* \to AB$

$* \to AC$

$* \to SC$

m[3,5]

b c c (3 4 5)

b | cc (3 | 4 5) → B ϕ

bc | c (3 4 | 5) → B C

$* \to BC$

G : $S \to AB$

$A \to AB \mid AA \mid a$

$B \to BC \mid b$

$C \to c$

$w = aabcc$ (positions 1 2 3 4 5)

i \ j	1	2	3	4	5
1	A	A	A,S	A,S	
2	—	A	A,S	A,S	S,A
3	—	—	B	B	B
4	—	—	—	C	ϕ
5	—	—	—	—	C

m[1,4]

a a b c

1 | 2 3 4

a | a b c

A   A,S

$* \to AA$

$* \to AS$

1 2 | 3 4

a a | b c

A   B

$* \to AB$

1 2 3 | 4

a a b | c

A,S   C

$* \to AC$

$* \to SC$

m[2,5]

2 3 4 5

a b c c

2 | 3 4 5

a | b c c

A   B

$* \to AB$

2 3 | 4 5

a b | c c

A,S ϕ

2 3 4 | 5

a b c | c

A,S   C

$* \to AC$

$* \to SC$

$G:\ S \rightarrow AB$

$A \rightarrow AB \mid AA \mid a$

$B \rightarrow BC \mid b$

$C \rightarrow c$

$w = aabcc$ (positions 1 2 3 4 5)

i \ j	1	2	3	4	5
1	A	A	A,S	A,S	A,S
2	—	A	A,S	S,A	A,S
3	—	—	B	B	B
4	—	—	—	C	ϕ
5	—	—	—	—	c

m[1,5]

aabcc

1 2 3 4 5
a | abcc
A   A,S

✗ $\rightarrow$ AA
✗ $\rightarrow$ AS

1 2 3 4 5
aa | bcc
A   B

$\rightarrow$ AB

1 2 3 4 5
aab | cc
A,S ϕ

1 2 3 4 5
aabc | c
A,S   c

✗ $\rightarrow$ AC
✗ $\rightarrow$ SC

In the table[1, n], If we can find the starting variable, then $w = w_1 w_2 \ldots\ldots w_n$ can be generated by G.

In m[1, 5] we have S

$\therefore$ aabcc can be generated by G.